

PEM-CAN: Parameter-Efficient Multimodal Diabetic Neuropathy Screening

Orthogonal Low-Rank Subspace Cross-Attention over Retinal Imaging & Wearable Biosignals

1. MULTIMODAL SENSORS



Retinal Oculomics

Cumulative Microvascular State

- High-resolution fundus photography
- Arteriolar narrowing & capillary rarefaction
- Spatial tokens: $M = 576$



Wearable Telemetry

Dynamic Autonomic Excursions

- 7-day CGM (5-min, $T = 2,016$ steps)
- 5-channel actigraphy: HRV, sleep, stress
- Multiscale Conv1D tokens: $L = 252$

Clinical Challenge

Visual dominance & quadratic attention blowup

2. PEM-CAN SUBSPACE ADAPTERS



Frozen ViT-B/16

86.6M Base Params
(Zero Backprop)



Temporal Conv

Multiscale Tokenizer
($L = 252$ steps)



Stiefel LoRA

$\|AA^T - I_r\|_F^2$ Penalty
Rank $r = 8$ Subspace
98.3k Adapter Weights

Bidirectional Cross-Attn

$S(Z_v, Z_s)$ & $S(Z_s, Z_v)$

- Linear Complexity: $O(M \cdot L)$
- Prevents Modality Collapse
- Zero-Overhead Inference
- Adapters fold into W_0

$L_{total} = L_{focal} + L_{orth} + L_{align}$

Multi-Task Head

DAN (Autonomic) | DR (Retina)
CKD (Renal) | DPN (Sensory)

3. BENCHMARK COHORT



AI-READI T2D Protocol

$N = 2,840$ Patients

- Statistically mapped to NIH atlas
- All confirmed Type 2 Diabetes
- Stratified 5-Fold Cross-Validation
- White (60%), Black (25%), Lat (15%)



Adjudicated Targets

Clinically Validated Labels

- DAN (Autonomic): 22.5% Prev.
- DR (Retinopathy): 18.2% Prev.
- CKD (Renal): 14.1% Prev.
- DPN (Peripheral): 26.8% Prev.

4. QUANTIFIED PERFORMANCE



0.934 AUROC

90.5% Accuracy | 0.803 F1

- Oracle Ceiling: 0.957
- ECE: 0.038



1.75% Weights (2.6M)

8.4 GB VRAM (vs. 32.4 GB)

- 0.861 AUROC under 50% sensor loss



20.2 ms Edge Latency

NVIDIA Jetson Orin Nano (1.18 GB)

- 50-Node FL: 98.25% less bandwidth